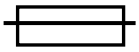
11 PLUS PREPARATION

NON-VERBAL REASONING PAPER

Name:	
Date of Birth:	
Today's Date:	

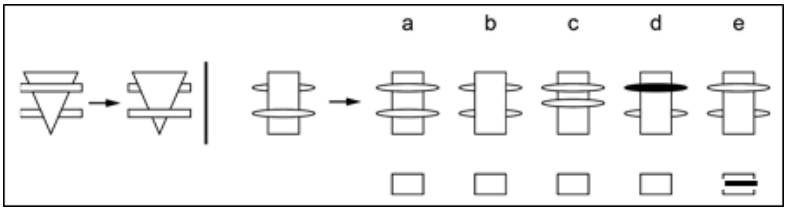
READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 25 minutes to complete the paper.
- Mark your answer like this: 
- This paper features a total of 39 questions from 13 different question types. An example is given at the start of each question type.
- Each question is worth one mark. If you can't answer a question, move on.
- If you finish early check your answers for mistakes.
- Once the test has started you may not ask for help.

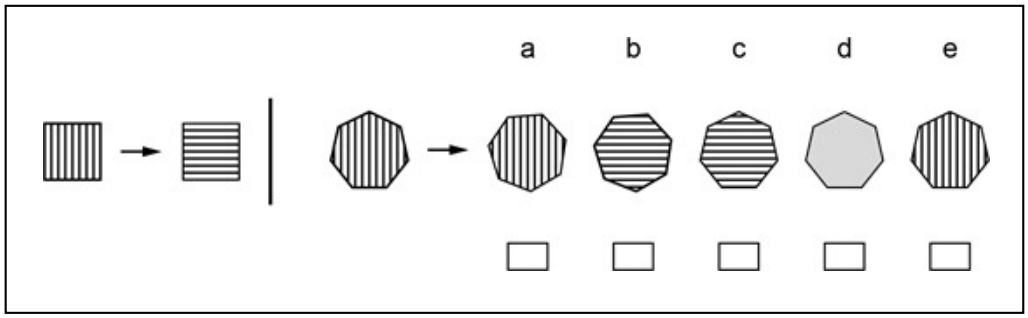
1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

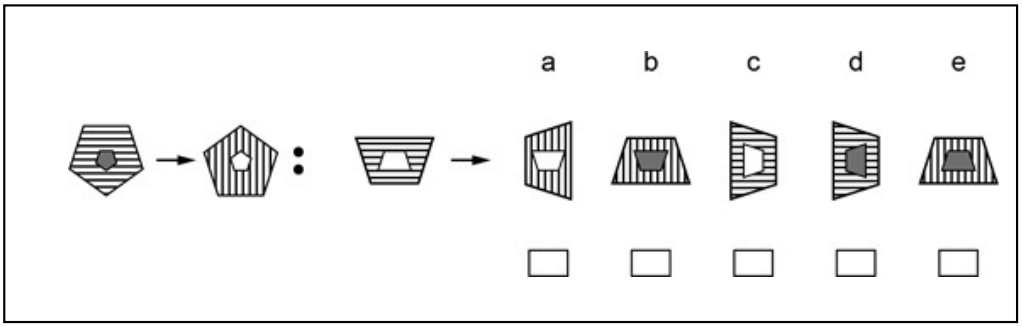
Example



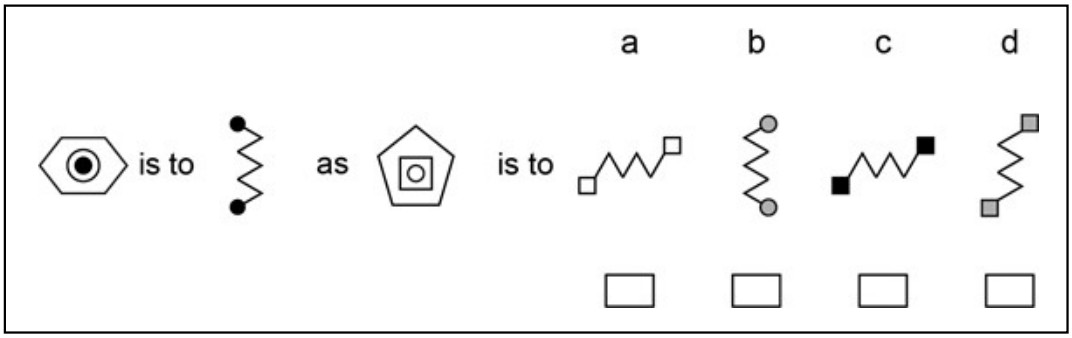
1



2



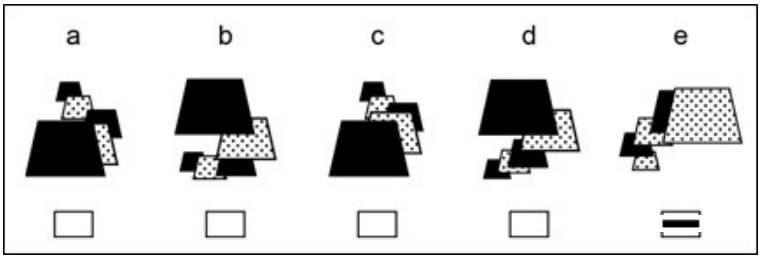
3



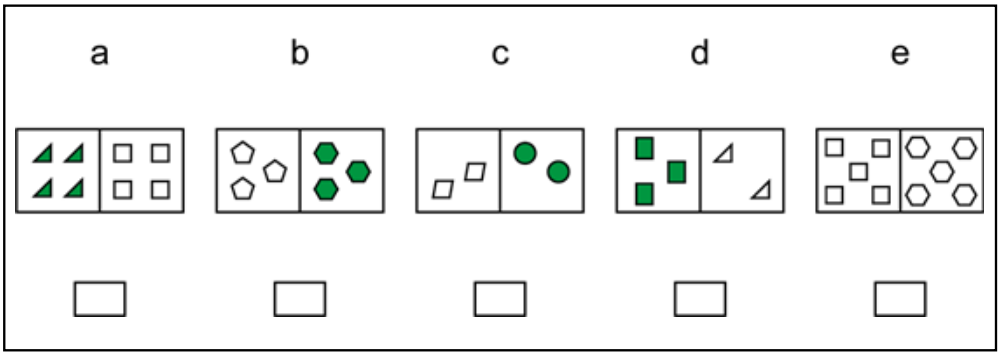
2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

Example



1



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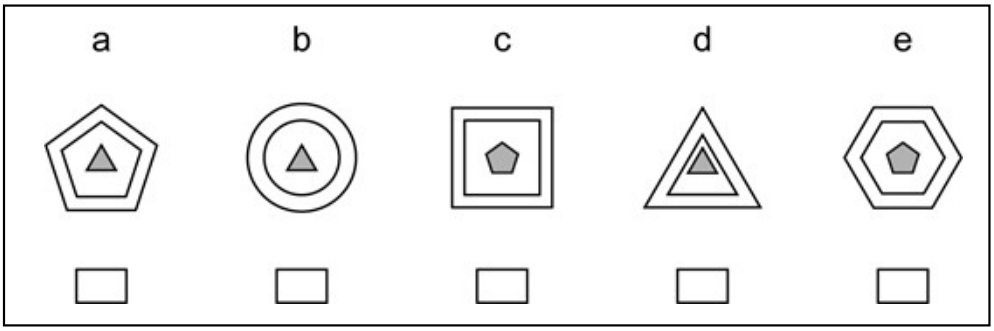
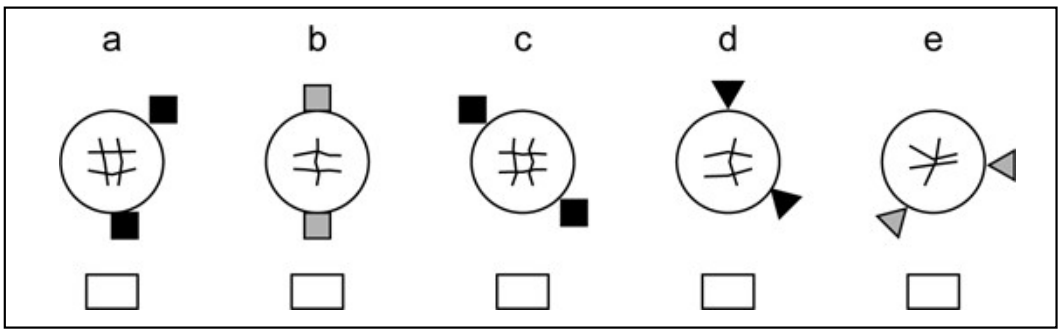


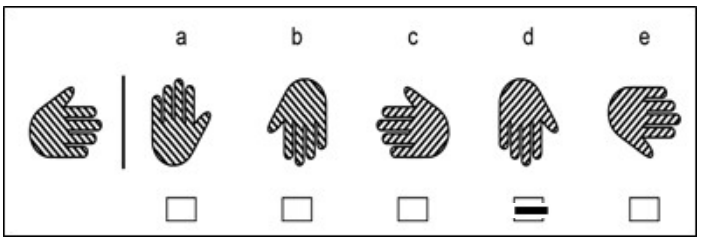
3



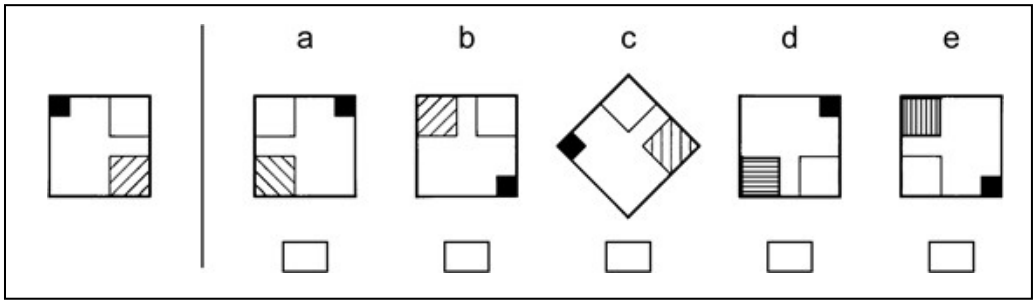
3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

Example



1



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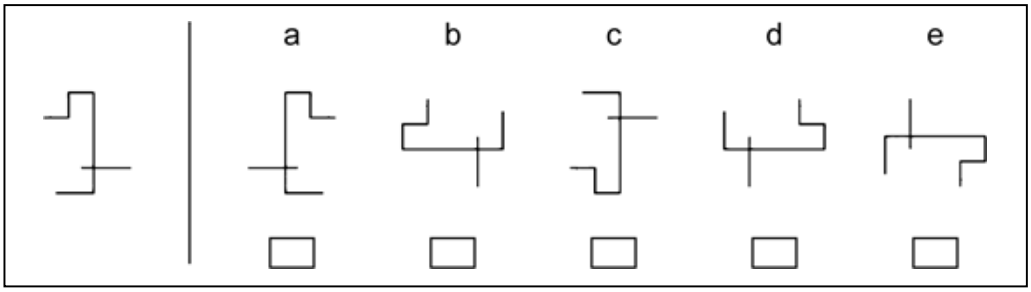

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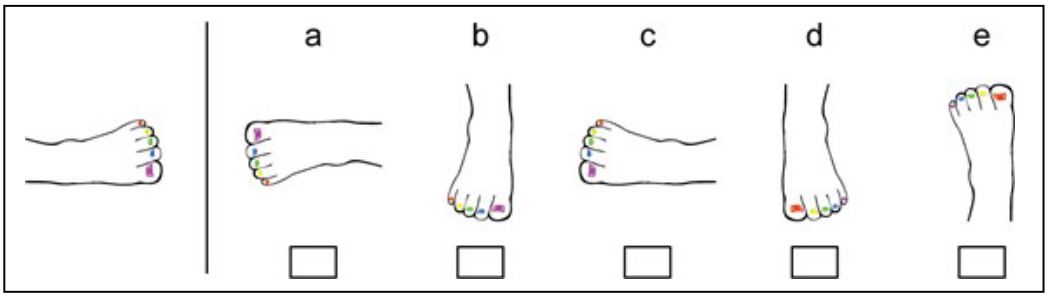
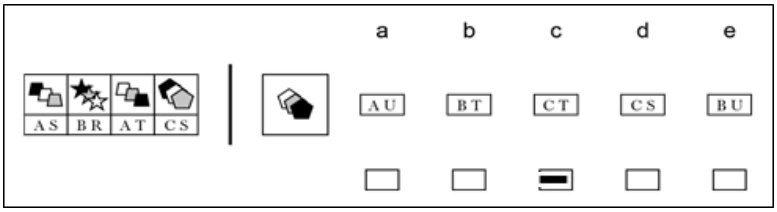


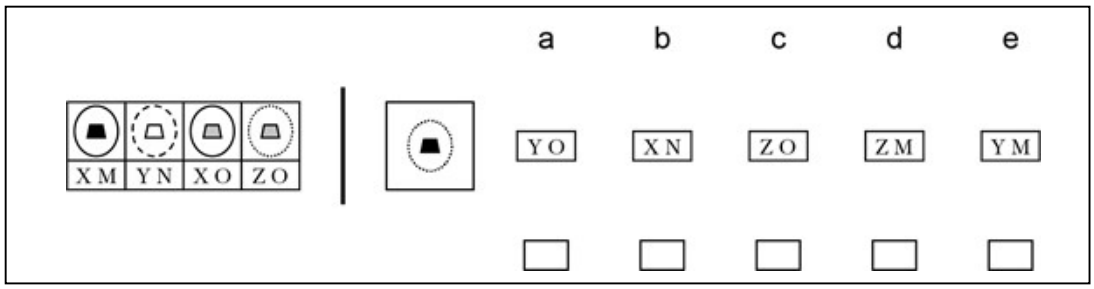
4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

Example



1



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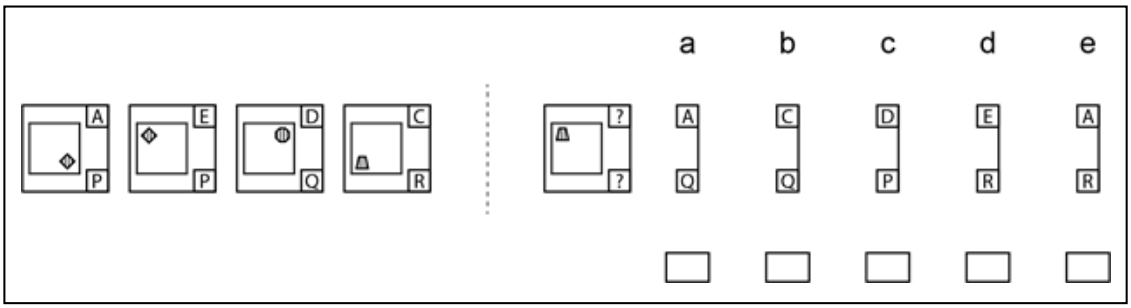

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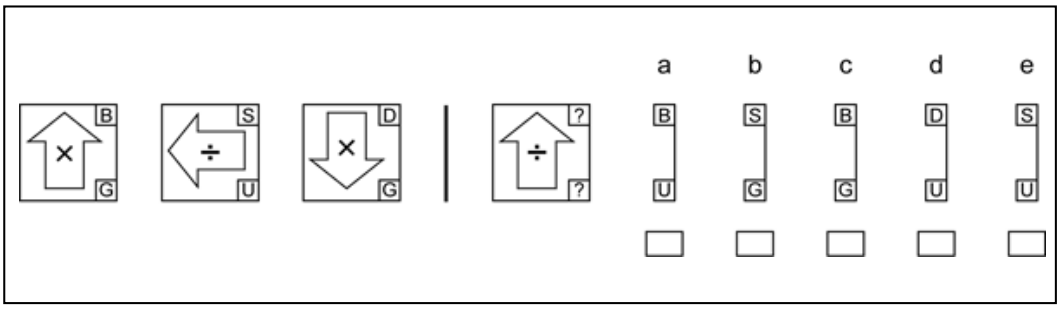
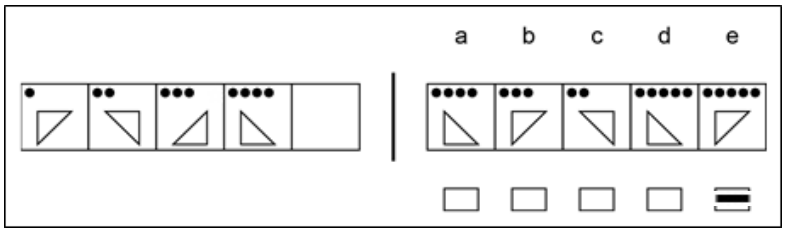


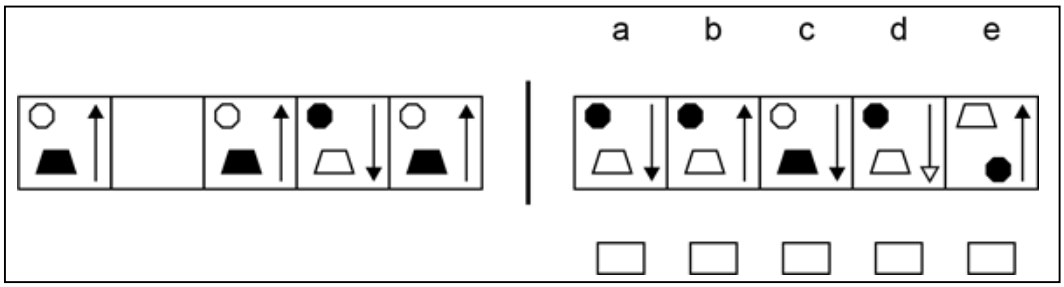
5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

Example



1



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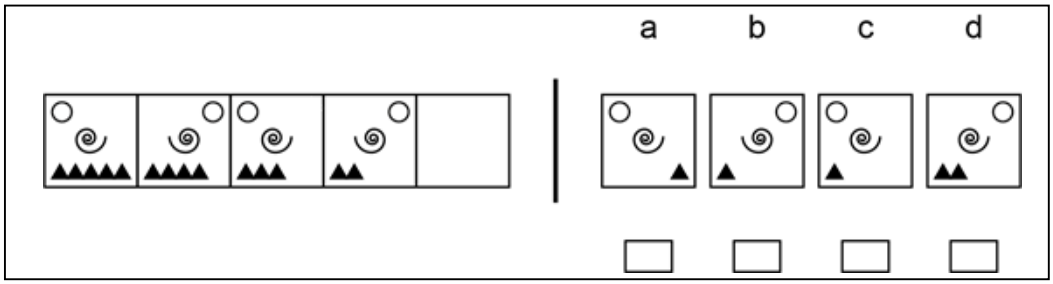

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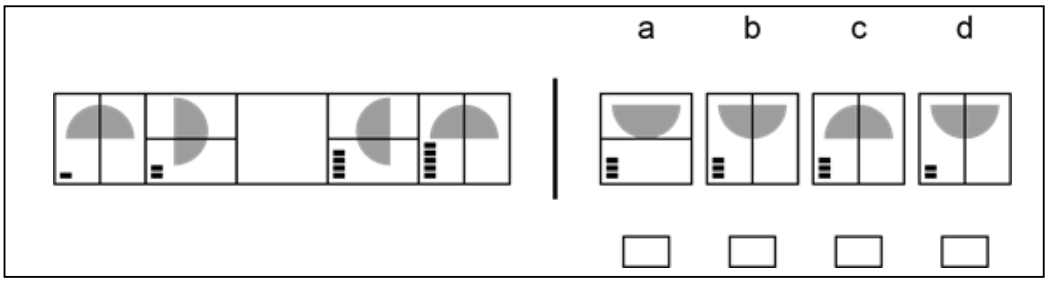
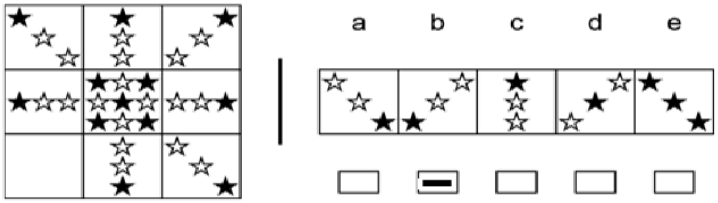


6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

Example



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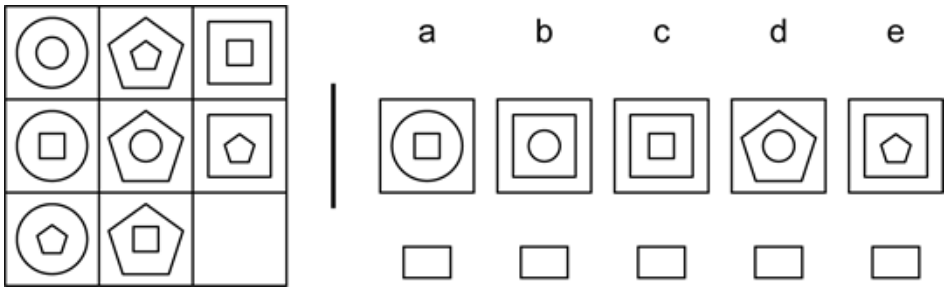

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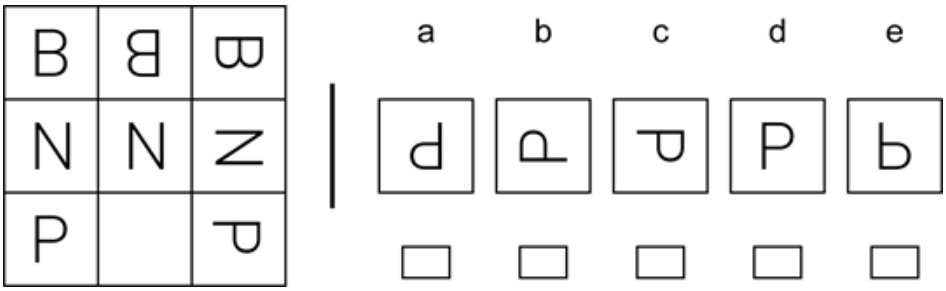

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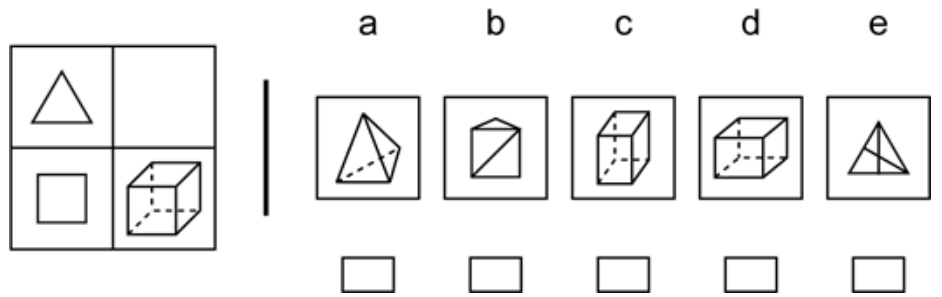
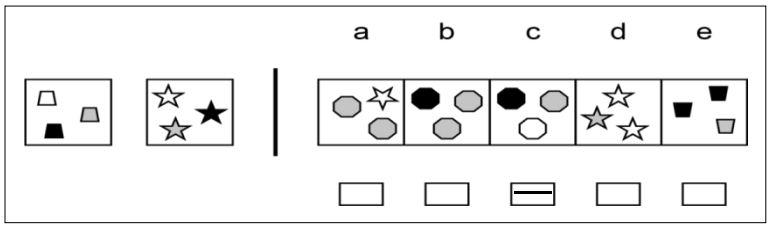


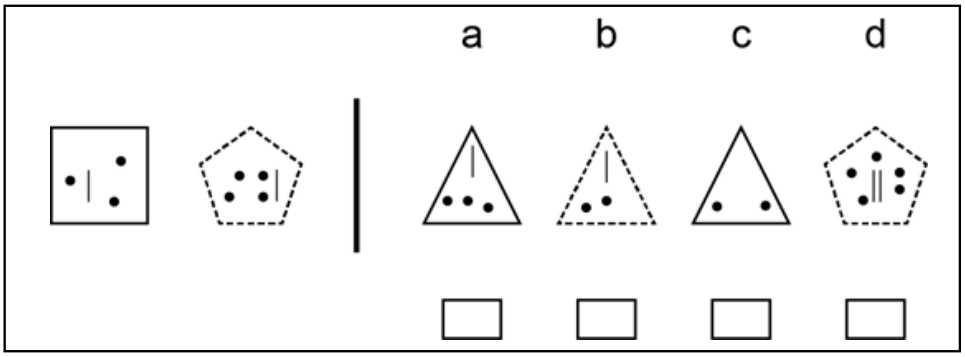
7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

Example



1



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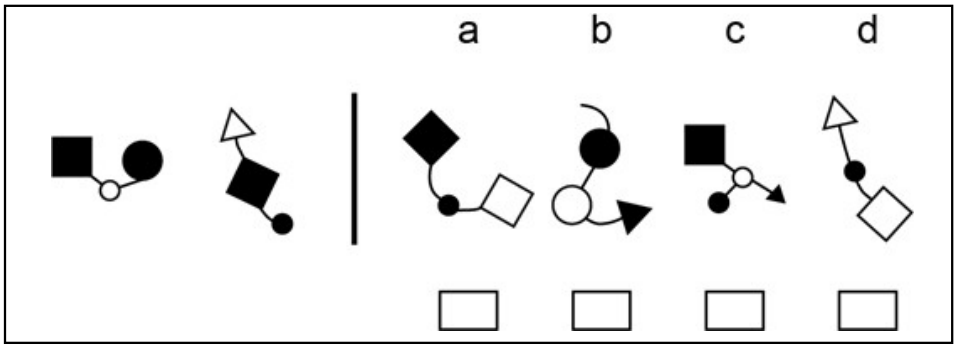
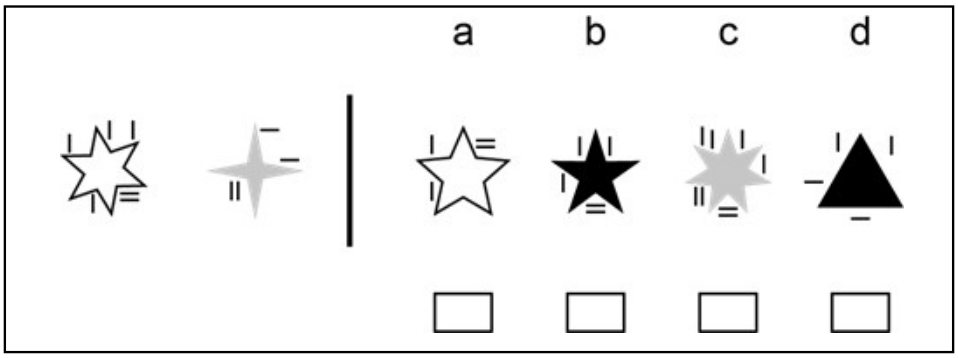


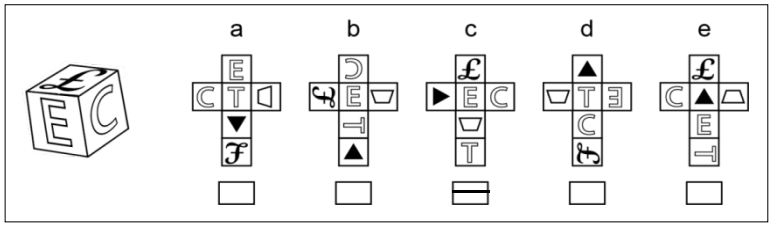
3



8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

Example



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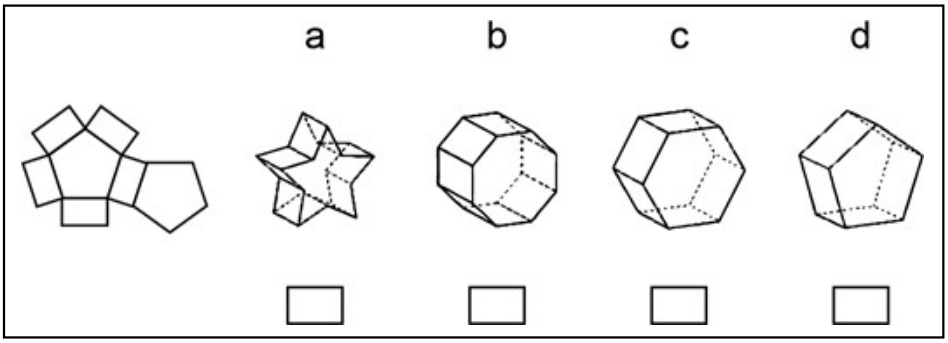
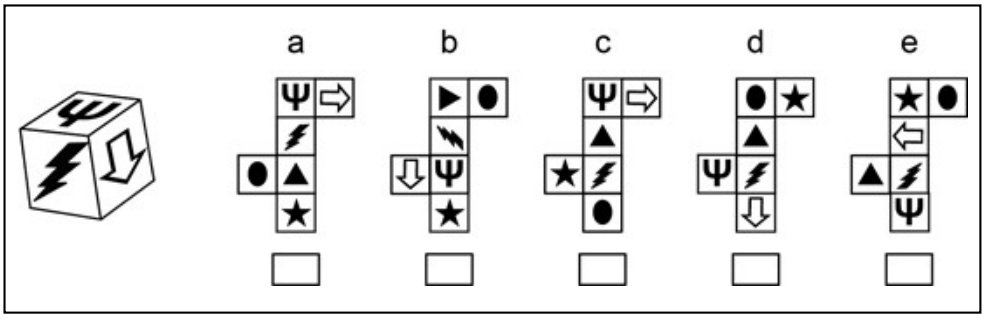


2



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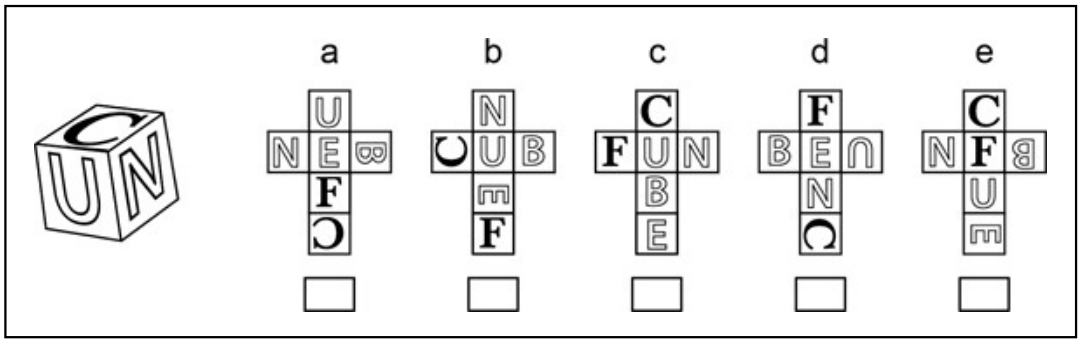
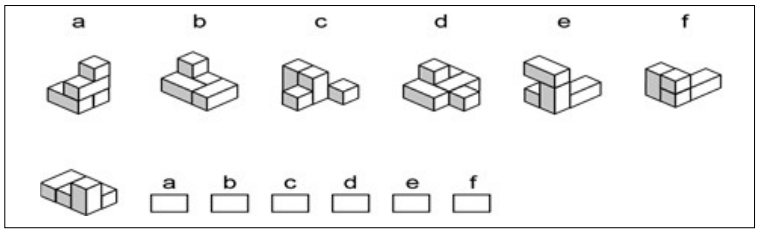


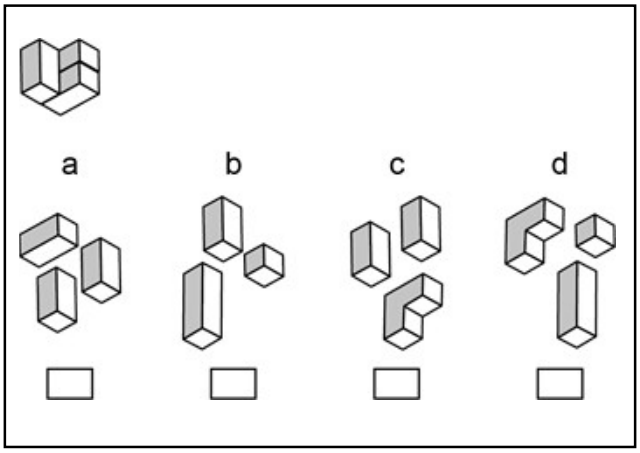
9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

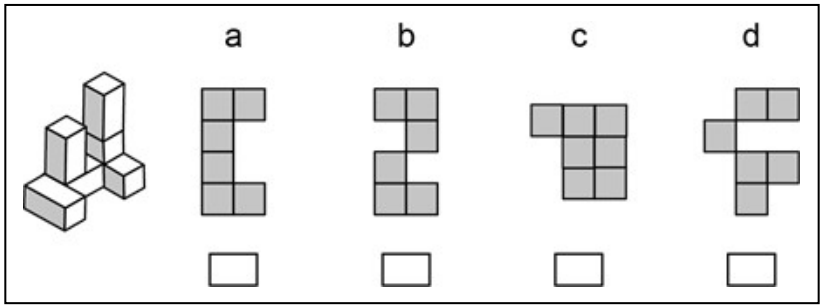
Example



1



2



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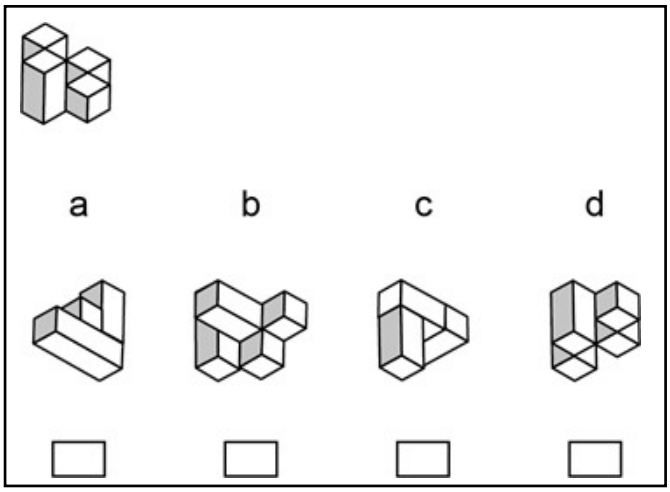
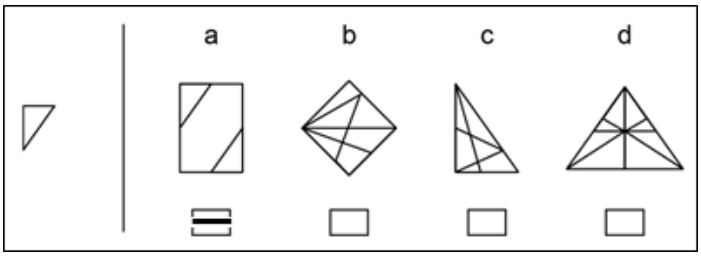


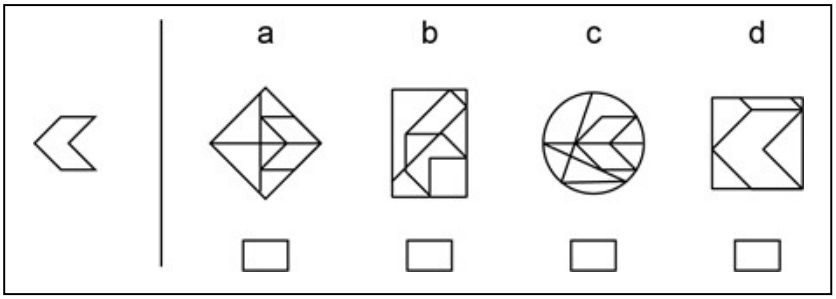
10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

Example



1



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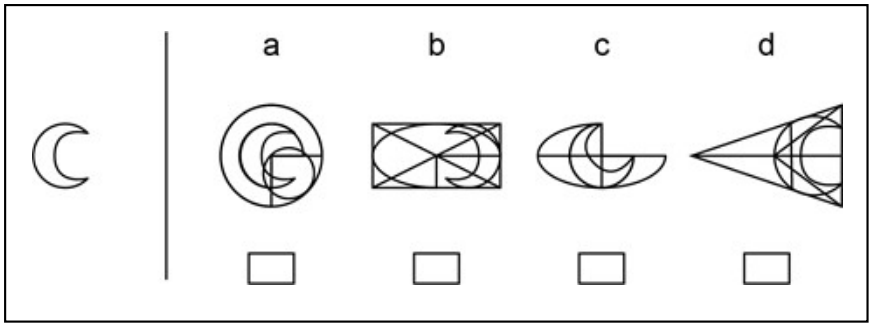

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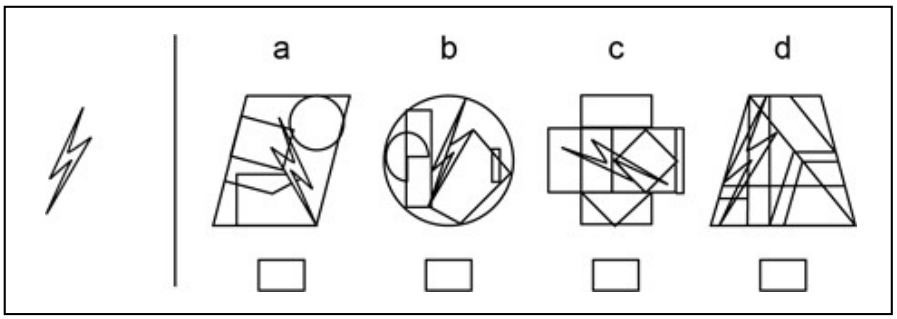
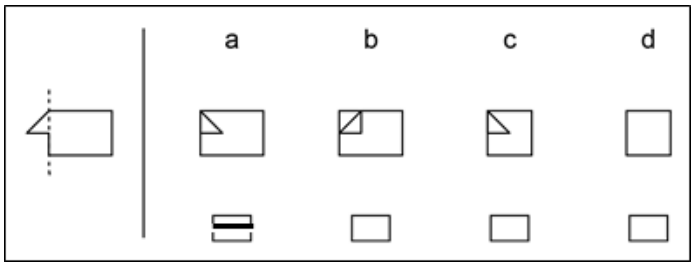


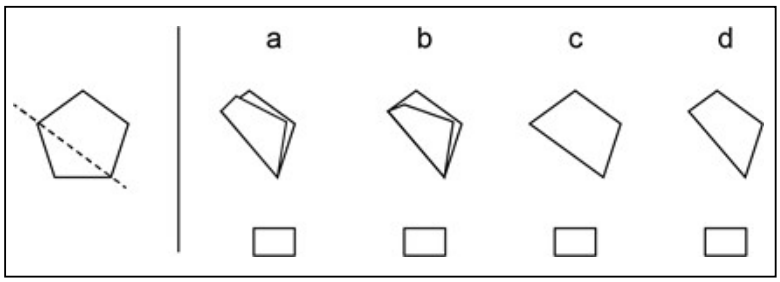
11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

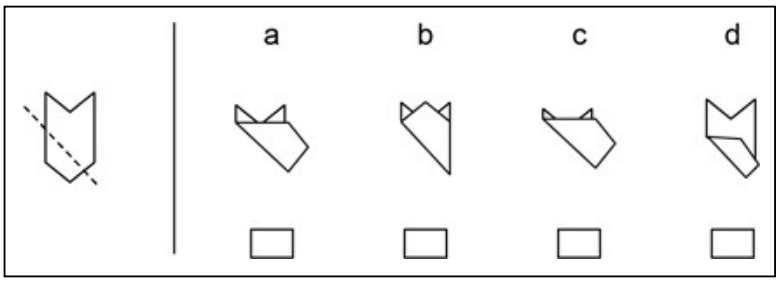
Example



1



2



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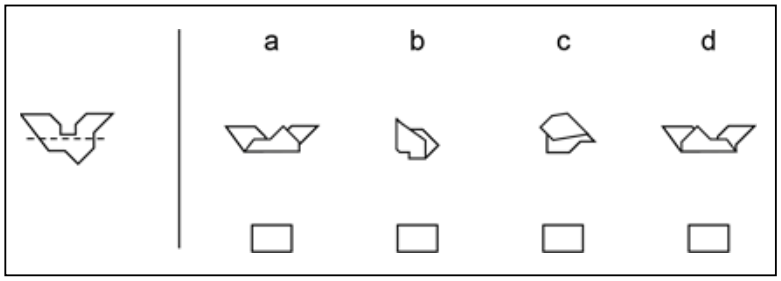
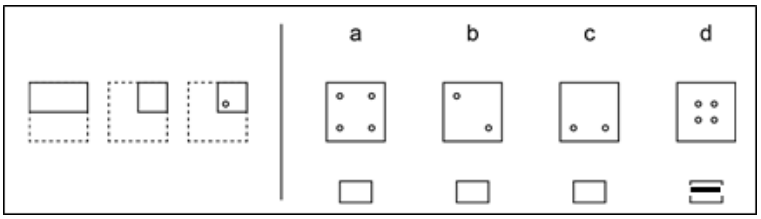


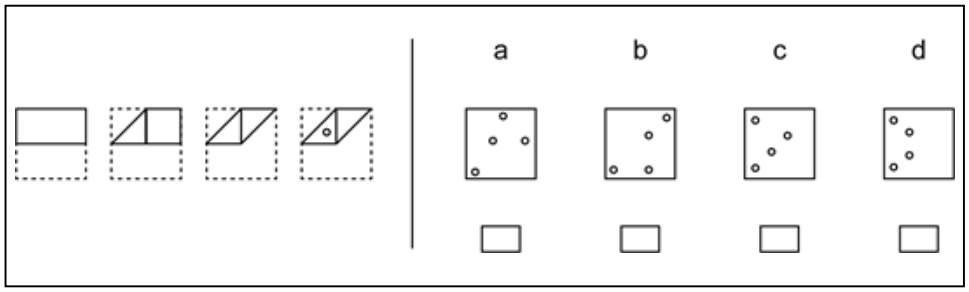
12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

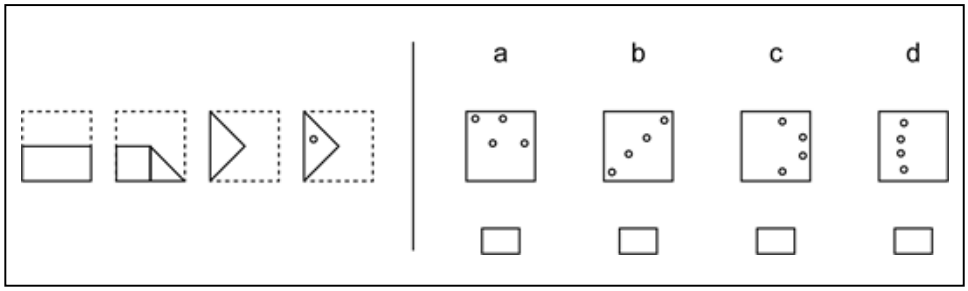
Example



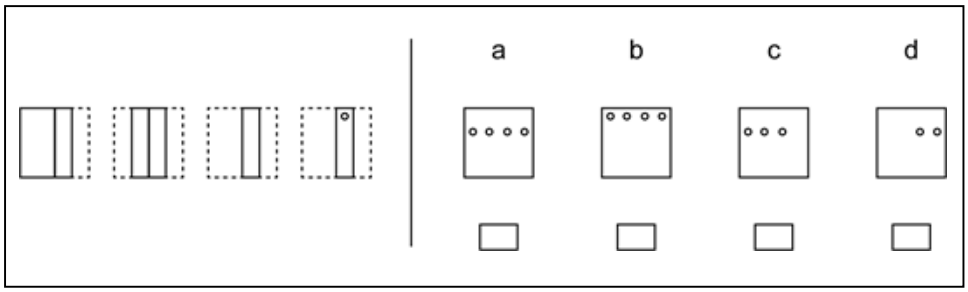
1



2



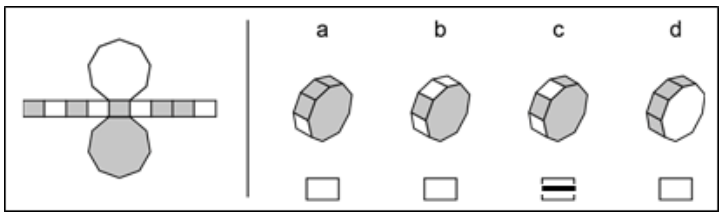
3



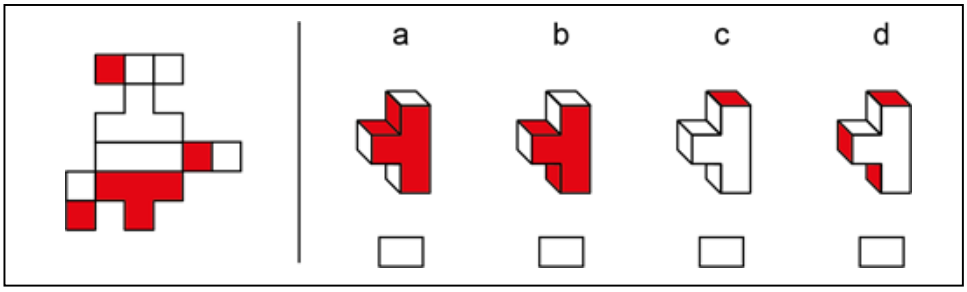
13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

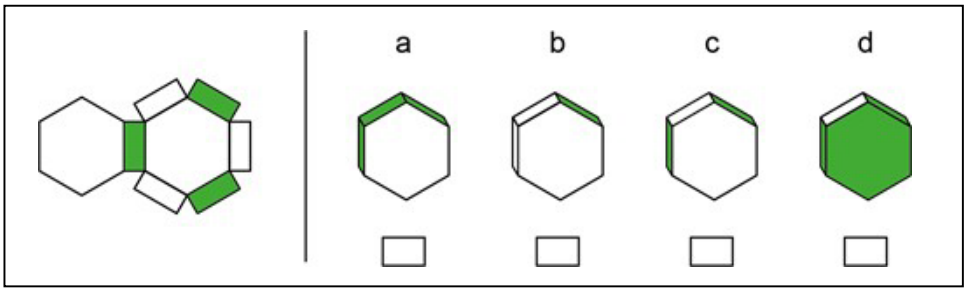
Example



1



2



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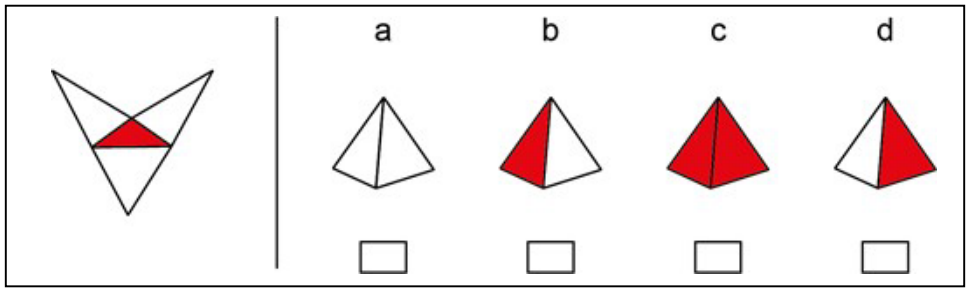


Non-Verbal Reasoning

Like Shapes

1. **C** - (The shape simply changes from vertical lines to horizontal.)
2. **B** - (Both the outer and inner shape are rotated 180 degrees. The horizontal lines in the outer shape are changed to vertical lines and the inner shape changes colour.)
3. **A** – (By analysing the example you can work out that the number of sides of the shape represents the number of lines, the middle shape represents the type of shape at the end of the lines and the inner shape represents the colour of the shape at the end of the lines. Therefore the answer must be two white squares connected by 5 lines.)

Odd One Out

1. **D** – (All options have the same number of shapes in both squares apart from D, which has three shapes in the left square and two in the right square.)
2. **D** – (All options have the innermost shape in the centre of the middle shape apart from D, which has the innermost shape towards the top of the middle shape.)
3. **A** – (All options have two shapes touching the circumference of the circle in the same way apart from A. The lines in the middle of the shapes are not relevant.)

Rotations

1. **C** – (C is a rotation of approximately 45 degrees anti-clockwise. Watch out for incorrect options such as D and E that look like they have been rotated correctly but for the shading in one of the squares.)
2. **D** – (D is a rotation of 90 degrees clockwise.)
3. **A** – (A is a rotation of 180 degrees. Make sure the toes are correctly painted!)

Code Breaker

1. **D** - (You can work out that X is the solid circle around the trapezium as there are two of them. Therefore this means that Y is the dashed circle and Z is the dotted circle. There are two Os in the second letter, so this must mean that O stands for the light shading of the trapezium. The solid black shading is therefore M. So the final answer is ZM.)
2. **D** - (You can work out that P stands for the small diamond shape as there are two of them. Therefore, Q stands for the octagon and the R stands for the trapezium. All of the top letters are different and the only thing different between all of the squares is the position of the small shape within the square. So, E stands for the shape being in the top left. Therefore the final answer is ER.)
3. **A** – (You can work out that G stands for a multiplication sign in the middle of the shape as there are two of them. U therefore stands for a division sign in the middle of the shape. Despite all the top letters being different, you can deduce that they represent the direction that the arrow is pointing as this is the only other difference between the three shapes. Therefore the answer is BU.)

Missing Sequence

1. **A** - (The squares keep alternating with three changes each time. The arrow changes direction from up to down or down to up, the trapezium changes colour from black to white or white to black and the octagon changes from white to black or black to white. The answer must therefore be A.)
2. **C** – (As you go through the sequence three changes occur: the number of triangles at the bottom decreases by one each time, the white circle alternates between being in the top left and top right corner, and the spiral in the centre alternates between pointing towards the right and pointing towards the left.)
3. **B** – (As you go through the sequence three changes occur: the number of dashes in the bottom left corner increases by one each time, the shaded semi-circle rotates by 90 degrees clockwise each time, and the solid line alternates between being vertical and horizontal.)

Missing Square

1. **B** – (The outer shapes remain in the same pattern throughout the grid, however as you go from left to right in each row the internal shape alternates between a circle, a pentagon and a square.)
2. **A** – (The letters in the middle column are a 180 degree rotation of the letters in the left column. The letters in the right column are then a 90 degree anti-clockwise rotation of the letters in the middle column.)
3. **A** – (The shapes in the right hand column are a 3D version of the respective 2D shape in the left hand column.)

Most Like

1. **B** – (The number of lines and dots inside the shape add to make the number of sides of the outer shape.)
2. **A** – (Two black shapes and one white shape are connected by two lines.)
3. **B** – (The number of lines around the star corresponds to the number of points it has.)

Nets

1. **D** - (You can see that the net is made up from two pentagons and five rectangles. The only shape that is made up from two pentagons and five rectangles is D, therefore the answer is D.)
2. **A** – (Using the respective position in relation to one another that you are given in the cube, you can see that the arrow must be able to fold around so that when it is to the right of the lightning bolt it is pointing down. The only option that satisfies this is A.)
3. **C** – (You can see from the cube that the letter C must be above the U and the N must be to the right of the N in the correct orientation. C is the only answer that satisfies this. Notice that the correct net also spells the word FUN!)

3D Shapes

1. **A** – (Looking at the 3D shape you can tell that it must be made up from three blocks of equal size. A is the only option that satisfies this.)
2. **A** – (On first sight this question appears quite tricky, however you can quickly dismiss B and C as incorrect answers because from the angle you can see the 3D shape from you know there must be at least two squares coming from the same side of the main shape. The choice is then between A and D. You can see on the 3D shape that at least a solid line of length four blocks will be at the base of the shape, therefore D cannot be correct.)
3. **D** – (The shape has been rotated so that it is upside down.)

Hidden Shape

1. **C** –

2. **A** –

3. **B** –

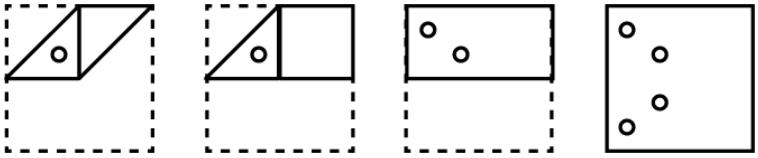
Fold Along the Dotted Line

1. **C** – (The left portion will fold down behind the right portion so that it cannot be seen. Notice that in the other answer options the fold line would be at the wrong angle.)
2. **D** – (The left portion will fold up over the top of the right portion.)

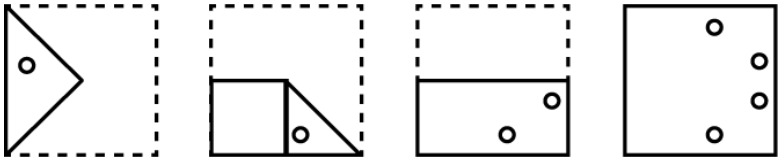
3. **A** – (The bottom half will fold up over the top half so that the wider parts of the top half are still visible underneath.)

Hole Punch

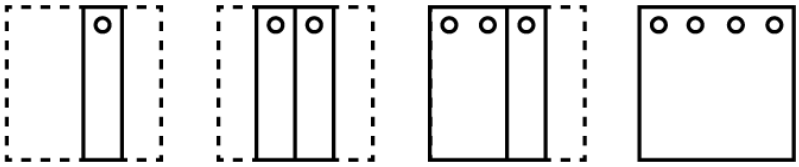
1. **D** –



2. **C** –



3. **B** –



Shaded Nets

1. **A** – (Options C and D can be dismissed fairly quickly as incorrect answers as the shading pattern of the squares is impossible based on the net. Therefore the answer is between A and B. You can work out that the answer has to be A because in order for the shape to appear in that orientation the net must be folded down and the resulting shape stood on its end. Therefore the two red squares must appear at the top.)
2. **C** – (You can see from the net that both hexagonal faces must be white, therefore D cannot be correct. You can also see that the rectangular faces must alternate white, green, white etc. Option C is the only shape that satisfies this.)
3. **A** – (You can see from the net that only the small triangular base, which is not visible in any of the answer options, can be red. All other triangular faces must be white. Therefore A must be the correct answer.)